



DEALER BEST PRACTICE SERIES

Technician Labor
Requirements Guide for
Greenfield Mine Sites

Application Maintenance Site Management Component Rebuild Component Life Management Safety MARC Management

Technician Labor Requirements Guide for Greenfield Mine Sites0

1.0 Introduction1

2.0 Best Practice Description.....1

3.0 Implementation Steps2

4.0 Results.....3

5.0 Resources Required3

6.0 Supporting Attachments / References3

7.0 Related Best Practices3

8.0 Acknowledgements.....3

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1.0 Introduction

This Best Practice provides a technician labor requirement guide to dealers maintaining a fleet of Caterpillar machines at a Greenfield mine site. It offers suggestions on how to calculate manpower requirements to a dealer that has not otherwise provided service in a mining application.

Attention should be given to adequate manpower as it helps ensure early success and an improved customer relationship. Management and technician structure should be developed based on the needs of the customer. This structure can range from just on-site technicians to a full maintenance department with project managers, planners, fleet analysts and service billers.

Parts and Service departments should be involved in developing the staffing requirements for on-site personnel. A review of existing staffing should be considered if functions such as invoicing and warranty would be conducted off site.

2.0 Best Practice Description

This Best Practice will outline the logic that should be applied when determining the technician labor requirements for a dealer who is new to mining. This will ensure that the technician workforce accurately reflects the maintenance strategy of the dealership, the machine population, and that the dealer can deliver the availability and reliability the customer is demanding.

Listed are the criteria used in developing the labor requirements

- Annual utilization per machine
- Fleet size by model
- Number of shifts
- Scheduling
- Total days off per year
- Technician hours worked per day
- Technician efficiency / availability

Roles that are calculated include

- Shop technicians
- Field technicians
- Supervisors
- Schedulers / Planners

To determine labor requirements there must be an understanding of the performance metrics for mobile mining equipment. The *Performance Metrics for Mobile Mining Equipment* document that is attached in this Best Practice provides useful information on the Maintenance Ratio. The Maintenance Ratio is an indication of the amount of effort required to keep equipment in service as well as the efficiency with which labor is deployed and the effectiveness of the workforce. The Maintenance Ratio is calculated by dividing Maintenance and Repair Man-Hours by Operating Hours. The Maintenance Ratio could be lower for machines where the site performance is defined as excellent versus poor. It is not realistic to say that a Greenfield site will be operating as “excellent” immediately.

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3.0 Implementation Steps

The *Labor Analysis.xls* spreadsheet attached to this document should be considered a reference point for determining labor requirements. The values calculated in this spreadsheet should not be considered finite. It is not intended to replace sound maintenance planning procedures.

The information in this best practice should be adapted to local requirements. The values used in the attached Labor Analysis.xls spreadsheet are worldwide averages. Caution must be taken when assessing local abilities.

The following steps should be taken to determine labor requirements:

1. In the “**Man Hours**” tab enter
 - a. Days in a year
 - b. Days of leave etc.
 - c. Days in cycle
 - d. Days worked per cycle
 - e. Hours worked per day
 - f. Shifts worked per day
 - g. Efficiency Factor

The **Efficiency Factor** is the percentage of time the technician actually spends working. The factor can and should be adapted to fit local requirements.

2. In the “**Maintenance Ratios**” tab
 - a. Review maintenance ratios for each machine type
 - b. Modify maintenance ratios as necessary

The **Maintenance Ratios** are worldwide averages. They can and should be adapted to fit local requirements.

3. In the “**Calculations**” tab
 - a. Review the Ratio of total techs
 - b. Modify Ratio of total techs as necessary
 - c. Review the Ratio of management to total techs
 - d. Modify the Ratio of management to total techs as necessary

4. In the “**Fleet Entry**” tab enter
 - a. Each machine model by selecting it from the drop down box
 - b. Quantity of each machine model
 - c. Annual utilization of each machine model

5. In the “**Fleet List**” tab enter
 - a. Model
 - b. Serial Number
 - c. Planned Annual Usage
 - d. Start Hours

6. What to do if your machine is not listed in drop down box
 - a. In the “Maintenance Ratios” tab
 1. Change the name of one of the “other” machines to fit your need
 2. Enter the maintenance ratio of the new machine

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Greenfield Sites

DATE
2/20/2015

CHG
NO
00

NUMBER
0000-1.1-1001

- b. In the “Calculations” tab
 - 1. Change the name of the same “other” machine that was changed in the “Maintenance Ratios” tab
- c. In the “Fleet Entry” tab
 - 1. Unhide column “O”
 - 1. Highlight columns “N” and “P”
 - 2. “Right Click” while your cursor is in the highlighted columns
 - 3. Select Unhide
 - 4. Change the name of the same “other” machine that was changed in the “Maintenance Ratios” and “Calculations” tab
 - 2. Rehide column “O”
 - 1. Highlight column “O”
 - 2. “Right Click” while your cursor is in the highlighted columns
 - 3. Select Hide

4.0 Results

The *Labor Analysis.xls* spreadsheet provides a baseline number of shop and field technicians required to perform the maintenance on a Caterpillar fleet at a new mine site. It also gives a number of direct supervisors and schedulers / planners needed per the number of technicians. There is an assumption that equal amounts of work are done on each shift.

The numbers identified in this tool should be considered a reference point. The values calculated in this spreadsheet should not be considered finite. It is not intended to replace sound maintenance planning procedures.

5.0 Resources Required

People – an understanding of the required maintenance data is required

6.0 Supporting Attachments / References

Performance Metrics for Mobile Mining Equipment.pdf
 Greenfield Guidebook.pdf
 Labor Analysis.xls

7.0 Related Best Practices

NA

8.0 Acknowledgements

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